Does the Final Answer Measure the Agent? Outcome-Only vs. Trajectory-Aware vs. Human Evaluation of a Research-Assistant Agent

CS 6180 Foundations of Generative AI (Fall 2026) — Project Proposal

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1 Introduction

Problem. LLM agents that search, read and reason over several steps are usually scored only on whether the final answer matches a reference. This project measures *how much* that outcome-only score disagrees with trajectory-aware metrics (which score the search-and-read process) and with human judgment of the same runs, and whether the disagreement changes which model or configuration is judged "best".

Why it matters. A correct answer can come from a poor process: the model may recall it from pretraining, cite documents that do not support it, or succeed by luck on one run. A wrong answer can come from a sound search. If outcome-only evaluation hides these cases, leaderboards and outcome-only training rewards [12] optimize the wrong signal. Trajectory evaluation could make agents more *reliable*, *safer* and more *interpretable*, but its LLM judges have their own biases and costs.

Broader context. The work sits in *evaluation and benchmarking of generative-AI agents*, overlapping *alignment/safety* (grounded answers) and *LLM-as-a-judge validity*. Recent work argues that outcome-only evaluation is insufficient [1, 4], but not *how often* it misleads or *for which decisions*.

Planned study. A ReAct [13] research-assistant agent searches a frozen HotpotQA [14] corpus with gold evidence labels, using Qwen3 models (1.7B, 4B, 8B; thinking on/off) on free Kaggle GPUs, over 400 questions $\times$ 3 seeds plus closed-book and single-retrieval baselines. Each trajectory is scored (A) outcome-only, (B) trajectory-aware (evidence recall, citation grounding, redundant calls, LLM step judge), and (C) by blinded human scoring of 150 trajectories with reliability checks.

Impact. *Scientific:* measured disagreement rates with confidence intervals. *Practical:* which cheap signal best predicts human judgment per unit of compute. *Community:* public trajectories, judge outputs and human labels.

2 Background / Literature Review

AI Agents That Matter [2] and HAL [3]. Propose cost-controlled evaluation with accuracy–cost Pareto frontiers; HAL's 21,730 rollouts show more reasoning effort often *reduces* accuracy, and its log inspection reveals hidden behaviors. Strong evaluation hygiene. *Limitation:* quality is still task success; logs are read qualitatively, not scored.

TRACE [4]. The closest work: names the "high-score illusion" of Pass@1 and scores trajectories by efficiency and evidence grounding, exposing trade-offs accuracy hides. *Limitation:* its evaluator is not validated against humans, checked for bias, or tested across repeated runs.

AgentProcessBench [5]. 8,509 human step labels ($+1/0/-1$) on 1,000 tool-use trajectories; weak policies look better because they stop early. I reuse its ternary labels. *Limitation:* it evaluates judges, not outcome–process disagreement on the same runs.

TRAIL [6] and MAST [7]. Rigorous error taxonomies for agent traces (MAST: 14 failure modes, $\kappa = 0.88$). *Limitation:* annotation is costly (about 110 min/trace in TRAIL; six experts in MAST) and unrelated to outcome scores.

Judging LLM-as-a-Judge [8]. LLM judges match human agreement levels but show position, verbosity and self-preference bias; position swapping mitigates this. *Limitation:* tested on single-turn chat, not on agent trajectories, which LLM judges now routinely score [9, 10].

τ-bench [11]. pass^k (all k runs succeed) shows GPT-4o falling from over 60% to under 25% at $k = 8$; a simple, revealing metric. *Limitation:* still outcome-only: it shows *that* runs diverge, not *why*.

Search-R1 [12]. Trains search agents with RL using only exact-match reward on a reproducible local retriever, which I reuse. *Limitation:* this reward cannot tell retrieved answers from memorized ones.

Gap. No study applies outcome, trajectory and human evaluation to the *same* trajectories and measures how often they disagree, whether rankings change at matched cost, and how reliable each evaluator is.

3 Research Question

Main question: *On identical agent trajectories, how much do outcome-only and trajectory-aware evaluations disagree, does it change which configuration is judged best, and which automatic evaluator best tracks human judgment per unit of cost?* Hypotheses are pre-registered before full runs:

- **RQ1 (disagreement).** How often is the answer right but the process bad, or vice versa? *H1:* at least 10% of correct answers are ungrounded, mostly explained by memorization (compared against closed-book accuracy), and less often for larger models. *Measure:* 2×2 table of correctness vs. grounding.

- **RQ2 (rankings).** Do the three evaluations rank the 6 configurations alike? *H2:* outcome and human rankings differ at matched token cost (Kendall $\tau < 0.8$) [2].
- **RQ3 (validity).** Which automatic metric best predicts human scores, at what cost? *H3:* rule-based metrics (evidence recall, redundant calls) match an LLM step judge in Spearman ρ; LLM judges show measurable bias [8].
- **RQ4 (reliability).** *H4:* pass^3 is below 70% of pass^1 for every configuration [11]; seed-to-seed trajectory diffs show why.

The minimum deliverable is RQ1 and RQ4 on the 4B model.

Why worth answering / gap filled. It turns the accepted claim "final-answer accuracy is not enough" into effect sizes with confidence intervals and decision-level consequences, the measurement missing in Section 2. The design keeps every hypothesis falsifiable: paired questions, 3 seeds, a closed-book contamination control, and a human baseline with measured reliability (intra-rater re-test, external second rater for Cohen's κ). **Who benefits:** benchmark designers, practitioners choosing models from leaderboards, researchers training agents with outcome rewards, and builders of LLM judges.

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